The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.5.0 (prototype)
Log entry date	3 November 2026
Status	Released: deployed by the push of this release to the GitHub main branch (the commit titled Release 0.5.0), together with the 0.5.0 documents and this log. Netlify deploys the site automatically.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine, main branch (Release 0.5.0)
Previous version	0.4.5, 19 October 2026, commit 5515838 (see Spaghetti_Engine_Dev_Log_v0.4.5.docx)
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1. Purpose of This Log

This Development Log records the changes from version 0.4.5 (19 October 2026) to version 0.5.0 of The Spaghetti Engine, the first release of the roadmap theme of guided first use. Version 0.5.0 adds a library of named models in the browser, a Getting started guide, and a Basic/Advanced switch; removes the built-in example models; simplifies the Network tab's toolbar; makes arrows meet the boxes; and makes the Network tab and the Model editor much faster on large models.

The log documents what changed and why, how existing results are affected, how the changes were checked, and the deployment status.

2. Summary of Version 0.5.0

Area	Change
Model library	Several named models saved in the browser (open, new, rename, copy, delete); import adds a model; the 0.4.x slot moves in as "My model"; the Transition Pathways setup is kept per model.
Getting started guide	A card that walks through a first map in seven steps, ticked off from the model's content.
Basic / Advanced	Basic shows five analysis tabs and hides the scope, folding, out-of-scope display, classic layouts, and simulation settings.
Example models	Removed from the tool; kept as test fixtures.
Network toolbar	One row with Arrange and View menus and one structure picker; Force-directed removed; menus close on an outside click or Escape.
Arrows	Fixed-length arrowheads whose tip touches the target's border; endpoints kept off rounded corners; bowed lines aimed along their curve.
Speed	On a 150-concept model: adding a concept or undoing from about 4.8 s to 0.05 s, a drag step from 0.2 s to under 0.01 s, leaving the Network tab for the Model editor from about 1-1.5 s to 0.03-0.06 s.
Record of 0.4.5	Version 0.4.5 was pushed as commit 5515838 on 19 October 2026.

3. The Changes in Detail

3.1 Model library

- **Storage.** An index (spaghetti-engine:library:v1) lists the models with their name, size, and dates and remembers the open one; each model is stored under its own key (spaghetti-engine:model:<id>), written 0.8 s after a change and when the page is closed. A failed write shows "Not saved: use Export JSON" in the header.

- **Upgrade.** The first time 0.5.0 opens in a browser with a 0.4.x model (spaghetti-engine:autosave:v1), that model is copied in as "My model" with its Transition Pathways setup; the old slot is left in place.
- **Switching.** The open model is saved first; results, charts, scope, folding, selection, and undo history are reset for the model opened.
- **Import and export.** Import adds the file as a new model named after the file; Export JSON is named after the model. Building from a pasted matrix still replaces the open model, after confirmation.

3.2 Getting started guide

- **Steps.** Open the Network tab; add a concept; name it; add a second and connect them; set the relationship (a relationship selected, or a weight other than the default 0.5); sort into modules (optional); run (a Baseline Equilibrium or scenario result).
- **When shown.** On a first visit, not for a model brought over from 0.4.x; closed and reopened with Guide in the header; the state is remembered in the browser.

3.3 Basic and Advanced

- **Basic.** Model editor, Network, Influence Metrics, Baseline Equilibrium, Scenarios & Simulation, and About; no scope bar, folding, out-of-scope display, classic layouts, Advanced options, or module summary. Switching to Basic resets the scope, unfolds the modules, and moves off a hidden tab.
- **Default.** Basic for a new visitor, Advanced for a browser with a 0.4.x model.

3.4 Example models removed

- The water-energy-food, peat meadow, and energy transition examples are gone from the tool and from the code; a first visit opens an empty "My first model". They are kept in tests/fixtures/example-models.js, which the regression tests load.

3.5 Network tab

- **Toolbar.** Add concept, Undo, Redo, Arrange (smart layouts with a one-line description each; classic layouts in Advanced mode; Force-directed removed), View (style, lines, weights, overview map, out-of-scope display), a structure picker, search, Export PNG, and the panel button; the Inspector starts hidden on a narrow window and opens when a new concept needs its name.
- **Arrows.** The arrowhead length is set per relationship (9 + 3|w| pixels in Figure style, 11 + 5|w| in Editing style) instead of a multiple of the line width, and the line stops one tenth of that short of the border, so the tip touches it; a lone endpoint stays 12 pixels from a corner; a bowed line's free ends are re-aimed at its control point.
- **Speed.** The workbook's metric sheets are computed when the workbook is exported, not after every edit; a dragged card's position stays in the canvas until it is dropped; cards and lines are memoized and unchanged lists keep their identity; the 12 connection dots of a card are mounted only while usable; hover highlighting, and the zoom of labels, are done in CSS; routing and fan-out are keyed on the map's geometry and use a grid index; the Inspector is memoized; the Model editor draws its lists in steps and fills its concept pickers only when opened; the overview map is off by default.

4. Implementation Choices

Topic	Choice in the tool	Reason
Where models are kept	localStorage, one key per model plus an index.	Opening the list reads only the index; one damaged model cannot affect the others.
The 0.4.x slot	Copied, not moved.	An older version opened in the same browser still finds its model.
What the guide checks	The model's content, not clicks on the card.	The guide follows along however the model is built, including an import.
Basic and scope	Switching to Basic resets the scope and folding.	A scope that cannot be seen or changed must not silently restrict every analysis.
Drag and undo	The model changes once, on drop.	Writing each frame re-rendered the whole app 60 times a second; the drag is now one undo step from its starting position.

5. Effect on Existing Results

- No analysis result changes. A model from 0.4.x opens unchanged as "My model", in Advanced mode.
- Results reported with the former example models can no longer be reproduced in the tool directly; import the model from a JSON file, or ask for the fixture file.
- Version raised from 0.4.5 to 0.5.0 in package.json and package-lock.json.

6. Defects Found and Fixed

- **Undo after a drag.** Undo after dragging a card returned it only to where it was a frame before it was dropped, because history was committed after the moves; it now returns it to its starting position.
- **Hover and weight labels.** With the hover highlight in CSS, weight labels did not fade with their lines during development; fixed before release.

7. Verification

- **Tests:** tests/test_marginal_benefit_tp.cjs (37 checks), tests/test_transition_direction.cjs (37), tests/test_pathway_blank_inputs.cjs (5), and tests/test_pathway_expert_robustness.cjs (28) pass, now loading the former examples from tests/fixtures/example-models.js; tests/test_fcm_layout.mjs (57) passes.
- **In the app (production build, separate browser storage):** a first visit (Basic, guide open, empty model), an upgrade from a 0.4.x model ("My model", Advanced, Transition Pathways setup kept), every guide step, and every library action (new, open, rename, copy, delete with confirmation); drawing a relationship, dragging, and undo; arrow tips measured within 1.5 pixels of the target's border.

- **Timing on a 150-concept, 450-relationship model:** select 16-33 ms, drag step 6-7 ms, drop 30-50 ms, add a concept or undo 50-70 ms, starting-value slider 33-45 ms per step, pan and zoom about 1 ms, opening the Network tab 110-140 ms, leaving it for the Model editor 30-60 ms.
- **Build:** the production build completes without errors.

8. Deployment Status

- Deployed with the push of the release commit to main on 3 November 2026, together with the 0.5.0 documents and this log. Netlify rebuilds the site automatically.
- **Files changed:** src/SpaghettiEngine.jsx, src/AboutTab.jsx, src/modelLibrary.js (new), tests/fixtures/example-models.js (new), the four regression tests, package.json and package-lock.json, and README.md; with the documents, src/docs-manifest.json, public/sitemap.xml, and the three PDFs in public/docs, where the 0.5.0 versions replace those of 0.4.5.
- **Build note:** npm run build fails in the local OneDrive folder because of the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.
- **PDF note:** the PDFs were printed with Chrome (EDGE_PATH set to chrome.exe).

9. Prototype Scope and Known Limitations

- All limitations listed in the 0.4.0 to 0.4.5 logs still apply.
- Models are kept in one browser on one computer; the browser's storage (typically 5 MB per site) limits how many large models fit. Export JSON remains the way to keep or move a model.
- The User Manual and the Methodological Annex still do not describe the expert-defined sequences and the three robustness families of 0.4.4.
- On a 150-concept model each edit still costs about 30 ms in React Flow's own update of the cards.

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.5.0)	How to use every feature. Section 1.2 lists the changes of 0.5.0; Sections 3.5 to 3.7 describe your models, the guide, and Basic and Advanced; Section 4.1 the Network tab.
The Spaghetti Engine - Methodological Annex (version 0.5.0)	The methods. Section 14.5 now documents the former example models.
Spaghetti_Engine_Dev_Log_v0.4.5	The 0.4.5 release: Figure style and smart Auto Arrange.
This Development Log	What changed in 0.5.0 and the technical work behind it.

11. Next Steps

- Offer the former peat meadow example as a downloadable file, if wanted, so readers of the case study can open it.
- Describe the expert-defined sequences and the robustness families of 0.4.4 in the manual and the annex.
- Continue with the roadmap: version 0.6.0 (modelling made easier).